

HANDS-ON 7 - Angular Routing – Guards, Lazy Loading & Route Data

Aim:

To implement Angular Routing by configuring application routes, passing route parameters, implementing nested routes, lazy loading feature modules, and protecting routes using Route Guards.

Objective:

- Configure Angular application routing.
- Implement dynamic route parameters.
- Use query parameters in routing.
- Create nested routes.
- Implement lazy loading.
- Create and apply route guards.
- Prevent accidental navigation using CanDeactivate Guard.

Software Requirements:

- Visual Studio Code
- Node.js
- Angular CLI
- Angular 20
- Google Chrome

Components Created:

- Home Component
- Course List Component
- Course Detail Component
- Student Profile Component
- Courses Layout Component
- Not Found Component
- Enrollment Form Component
- Reactive Enrollment Form Component

Modules Created:

- Enrollment Feature Module
- Enrollment Routing Module

Guards Created:

- Auth Guard
- Unsaved Changes Guard

Services Used:

- CourseService
- EnrollmentService
- AuthService

Task 1 – Route Configuration, Parameters and Nested Routes:**Implemented Features**

- Configured application routes.
- Created Home route.
- Created Course List route.
- Created Course Detail route using route parameters.
- Created Student Profile route.
- Created Not Found route for invalid URLs.
- Read route parameters using ActivatedRoute.
- Navigated between components using Router.
- Implemented query parameters for searching courses.
- Implemented nested routing using Courses Layout Component.

Routes Configured:

Route	Component
/	HomeComponent
/courses	CourseListComponent
/courses/:id	CourseDetailComponent
/profile	StudentProfileComponent
/**	NotFoundComponent

Route Features Implemented:

- Route Parameters
- Query Parameters

- Nested Routes
- Router Navigation
- ActivatedRoute
- RouterOutlet

Task 2 – Lazy Loading and Route Guards:

Implemented Features

- Created Enrollment Feature Module.
- Configured Lazy Loading.
- Created Auth Guard.
- Protected Profile route.
- Protected Enrollment route.
- Created Unsaved Changes Guard.
- Displayed confirmation dialog before leaving unsaved forms.

Guards Implemented

Auth Guard

- Checks user login status.
- Allows authenticated users.
- Redirects unauthenticated users to Home page.

CanDeactivate Guard

- Checks whether the reactive form is modified.
- Displays confirmation dialog before navigation.
- Prevents accidental loss of data.

Lazy Loading

Lazy loading was implemented for the Enrollment Module.

The Enrollment JavaScript bundle is downloaded only when the user navigates to the Enrollment page, improving application performance.

Angular Concepts Used:

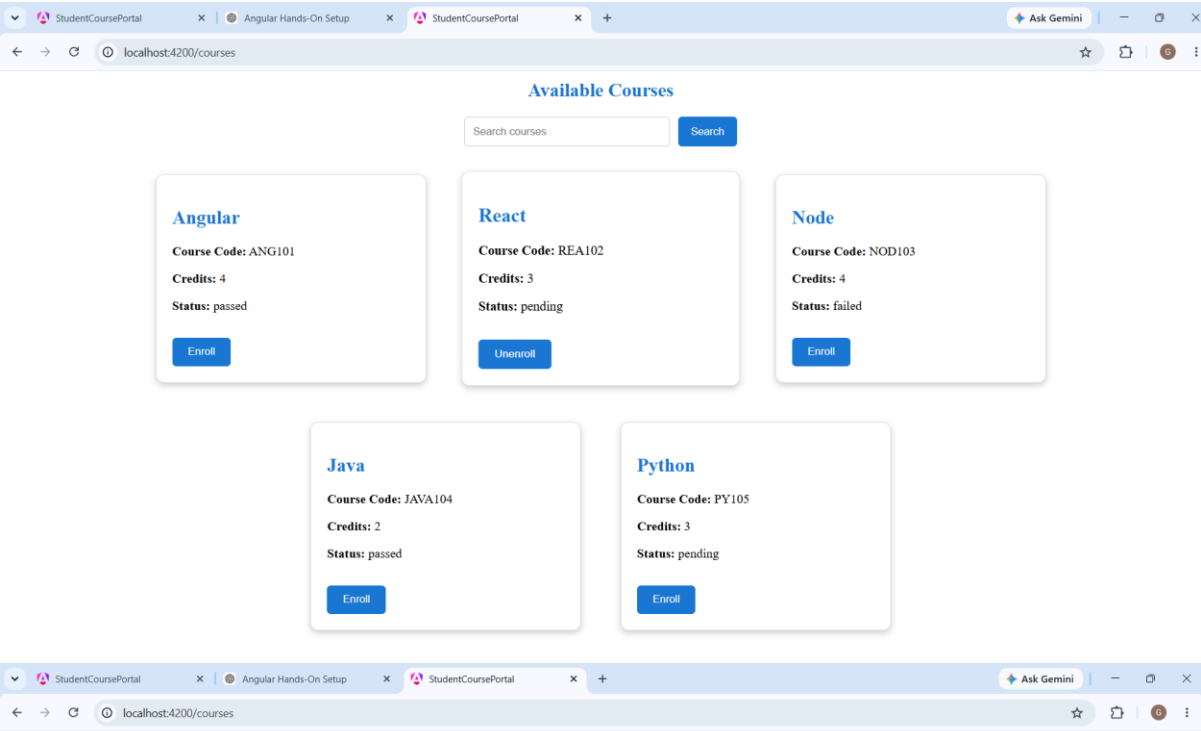
- RouterModule
- Routes
- Router
- ActivatedRoute

- Route Parameters
- Query Parameters
- Nested Routes
- Lazy Loading
- Auth Guard
- CanDeactivate Guard
- Feature Modules
- Router Outlet

Project Structure

```
src
|
|  └─ pages
|    |  └─ home
|    |  └─ course-list
|    |  └─ course-detail
|    |  └─ student-profile
|    |  └─ courses-layout
|    └─ not-found
|
|  └─ features
|    └─ enrollment
|      |  └─ enrollment.module.ts
|      └─ enrollment-routing.module.ts
|
|  └─ guards
|    |  └─ auth.guard.ts
|    └─ unsaved-changes.guard.ts
|
|  └─ services
|    |  └─ auth.service.ts
|    |  └─ course.service.ts
|    └─ enrollment.service.ts
|
└─ app.routes.ts
```

Outputs:



Student Course Portal

[Home](#)[Courses](#)[Enroll](#)[Profile](#)[Notification](#)

Available Courses

React

Search

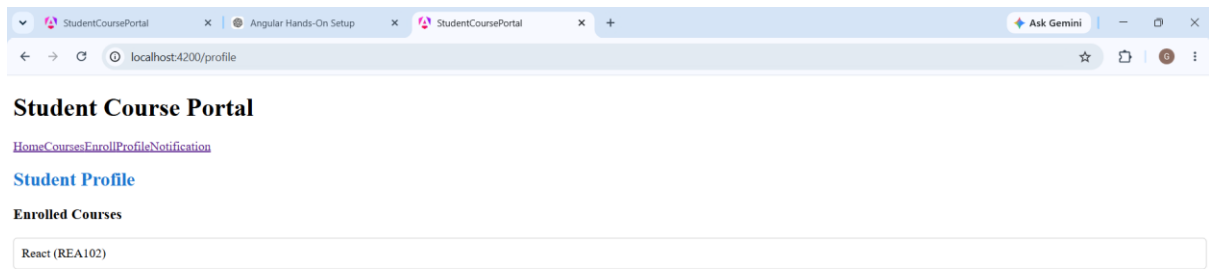
React

Course Code: REA102

Credits: 3

Status: pending

Enroll



Result

Successfully implemented Angular Routing using route configuration, dynamic route parameters, query parameters, nested routes, lazy loading, Auth Guard, CanActivate Guard, CanDeactivate Guard, and feature modules in the Student Course Portal application.